

College Ruled: A Pathfinding Approach to Generative Storytelling

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ABSTRACT

Automatic story generation remains a challenging task, with modern storytelling systems continuing to lag behind human-authored works in quality—either constrained in plot planning or confined to narrow story domains. While contemporary narrative planners and storytelling systems demonstrate improvements, static story generation systems still lag behind. We present College Ruled, a mixed-initiative storytelling system iterating on past works in static story generation such as Lebowitz’s UNIVERSE and Skorupski’s Wide Ruled re-imagination of the author-goal-based model of story generation. An effort to fold in the best qualities of modern storytelling systems into a static story generation context, College Ruled utilizes story waypoints, a drama management system, and causality weighting to guide plot fragment selection. This allows for authorial influence over how character relationships are shaped and provides agency over the dramatic arc of a spun tale. In conjunction with a stochastic modification to the A* algorithm, College Ruled offers a framework to be used alongside a library of manually authored plot fragments to produce rich variety in generated tales while meeting narrative goals. We describe the implementation of our system and then investigate its effectiveness in drama management and producing stories that meet author specifications.

CCS CONCEPTS

• Applied computing → Media arts.

KEYWORDS

Mixed-initiative Co-creativity, Authoring Tool, Procedural Content Generation, Story Sifting, Pathfinding

ACM Reference Format:

Nicholas Sloss Treynor and Joshua McCoy. 2024. College Ruled: A Pathfinding Approach to Generative Storytelling. In *Proceedings of the 19th International Conference on the Foundations of Digital Games (FDG 2024)*, May 21–24, 2024, Worcester, MA, USA. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3649921.3649994>

1 INTRODUCTION

Procedural content generation for narrative works and entertainment purposes has been a popular area of study, because even as storytelling remains an integral part of the human experience [18] [17], even modern approaches fall short of human-authored works,

frequently restricted to the abstract story representation level [22], requiring human-knowledge engineers to provide symbolic domain knowledge [12], or struggling in areas such as story coherency or creating a satisfying narrative arc [7]. Mixed authoring or “mixed-initiative” [8] [11] approaches sidestep this problem by including the author in the story generation process more heavily, to allow for the positive influence of insights and ideas not captured by fully autonomous narrative planners or storytelling systems.

Though generating static stories from author-supplied plot fragments is an established area of computational storytelling [9] [20], and so is the concept of drama-management within generated narratives, particularly in interactive formats [14] [17], directing the flow of a story within a static story generation system has not been explored to the same extent.

We aim to explore this gap with College Ruled, an iteration on the author-goal-based story generation system of Skorupski’s “Wide Ruled 2.0”, [20] itself a rational reconstruction of Lebowitz’s older UNIVERSE system [9], with alterations that allow for a higher degree of authorial control. College Ruled encodes author-goal worldstates—a configuration of the story world, characters, and their attributes—as story waypoints and then uses a pathfinding system to chart a course between them. Using a library of plot fragments—events that can occur within the story world, mutating attributes of the initial worldstate and characters to create a new worldstate while producing story text—the system explores a growing graph of possible stories from an initial starting point. By utilizing a pathfinding heuristic consisting of weighted worldstate character edit distance, drama curve fit, and causality incentives in conjunction with a non-deterministic modification to the A* algorithm, our system can produce multiple unique stories that meet an author’s narrative goals. By encoding drama levels into plot fragments and integrating them into our pathfinding heuristic, we iterate on existing narrative planners by constructing stories engineered to match a traditional tension arc [3]. This paper will describe our efforts to fold the best aspects of modern storytelling systems into a static story generation context, covering our pathfinding algorithm, heuristic, and system structure before evaluating our results.

2 RELATED WORKS

Procedural story generation and interactive narrative is a dense field filled with many approaches to these storytelling problems, ranging from classic AI techniques to ML[12] and NLP[22] methodologies. Those most relevant to our approach, however, are past static story generation systems and narrative planners.

In the early 1970s, developers started producing static storytelling systems like Meehan’s TALE-SPIN, a program that writes stories based on character interactions [15]. TALE-SPIN allows an

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FDG 2024, May 21–24, 2024, Worcester, MA, USA
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ACM ISBN 979-8-4007-0955-5/24/05
<https://doi.org/10.1145/3649921.3649994>

author to initialize a world with characters, locations, and character motivations, and then writes an entire narrative arc in which characters either fail or succeed at achieving their goals. Though a notable landmark in the history of computational storytelling, Tale-Spin is ultimately limited by its approach of merely simulating character interactions, rather than orienting itself around creating a world in which specific narrative desires can be achieved.

Lebowitz’s UNIVERSE [9] is perhaps one of the better-known static storytelling systems, attempting to facilitate the generation of stories with the dramatic arc of soap operas. It uses a rules-based system and rich character data to determine legal story events but tragically was left incomplete. Skorupski’s Wide Ruled 2.0 [20] iterates upon Lebowitz’s system, guided using the same principles, yet still is constrained by the techniques of UNIVERSE: It generates stories by churning randomly through available options in a hierarchical manner. Author goals are represented only through story-hierarchy rules, story generation is akin to taking a random walk on a graph configured by the author. While this results in coherent stories by ensuring a rigid sequence of interrelated events, and the quality of the produced text is high as it is manually authored, it is limited in the variety of output.

A more recent state-of-the-art system is STella, which explores story space using grounded knowledge representation [10]. STella uses a probabilistic system in conjunction with rules and author-defined goal curves (representing narrative tension, danger, etc.) to explore potential stories that grow out of a given starting point. However, once the starting parameters are provided, the author has little direct input in how the story progresses or ends. While an interesting artifact among static-story generation systems, we wish to bring a finer level of authorial control to the field with our system.

Lume [13], though not a static story generation system, requires mention due to its innovations in interactive story generation. Lume elevates the quality of spun stories through the use of recall, showcasing the consequences of prior actions in causing current events. Looking back through story history, Lume selects and builds scenes that refer back to earlier choices or events, elevating the quality of its story generation by ensuring that stories include causal sequences of events, improving story coherency. Lume is a hallmark of how far storytelling technology has come from the days of earlier systems like UNIVERSE and TALE-SPIN, which lack such features. While the system unfortunately remains proprietary and inaccessible, the concept of promoting sequences of causal events to improve story coherency is one we have borrowed for our system.

Glaive [21] is one of the most influential pieces of work in the context of narrative planners in recent years, utilizing a rigorous algorithmic approach to ensure a coherent flow of story from start to finish. It aims to generate a story that meets an author’s goals while being composed only of actions taken by characters that are sensible given the motivations of the characters. In doing so, Glaive is able to maintain a sense of believability in their generated stories. However, Glaive is still limited in its scope. It focuses on finding a valid, explainable path between starting and goal states, but may still lack some of the dramatic arc and flair of more finely crafted, human-authored tales. Additionally, as a narrative planner instead of a static story generation system, it focuses on finding a valid narrative solution, but doesn’t concern itself with generating

readable story text in the same way that systems like STella or Wide Ruled do. It is restricted to abstract story representation.

The proliferation of large language models also promises to be very interesting for storytelling systems in the near future. Works like TaleBrush [4] utilize LLMs in a mixed-initiative system to generate story text in response to user input. However, LLMs are not without fault. While such systems are capable of generating immediate continuation of human-written text, Kreminski et al. identify several areas of weakness in LLM-backed tools regarding storytelling: maintaining narrative consistency, constructing satisfying story arcs, managing reader experience, and iterating on authorial expressive intent [7]. Other investigations have also encountered issues with structural knowledge in stories generated by LLMs, resulting in incoherence when attempted to be used as an aid for novelists[2]. As a result, we believe there is still merit in exploring rules-based approaches further. That stated, we do see the value in utilizing LLMs, and believe they have promise in aiding with the generation of chunks of story to be used within a rules-based static story generation system. Our own efforts to leverage LLMs to generate plot fragments have proven fruitful, but this merits a discussion on its own and will remain outside the scope of this paper, though we are excited to discuss this in a future publication.

Ultimately, many advancements have been made in recent years in the field of algorithmic storytelling. We wish to bring such engagement and interest to static story generation systems, breathing new life into the older UNIVERSE-style system, and bringing it in line with more modern storytelling feats, while ensuring our methodology remains accessible. We hope to allow for a higher degree of authorial control over the dramatic arc of generated stories by integrating a drama management system in a style similar to that of STella’s goal curves [10], while also paying attention to causality in the manner that Lume [13] does to increase story coherency. Simultaneously, we will utilize a more rigorous algorithmic approach to story pathfinding, akin to that of Glaive [21] to allow for more complex narrative tasks to be asked of the static story generation style system than that which was previously possible to achieve using a system like Skorupski’s Wide Ruled [20].

3 TECHNICAL DESCRIPTION

3.1 Basic Objects

College Ruled is built out of a few basic objects, which interact with and encapsulate one another to determine what can take place in a story-space, and what should, in order to facilitate the author’s goals.

3.1.1 Worldstates. Worldstates are objects that contain a list of characters currently in the worldstate, the environments present in the world available for use, a log of what events have taken place preceding the current worldstate, a linked list history of prior worldstates, and a drama score. This information helps to encapsulate the story state at a particular point in the narrative, and what unfolded to get the story to that point. Authors can create initial worldstates and desired worldstates, or waypoints, that College Ruled will chart a path between. In Fig. 1, you can see the basic structure of a worldstate object and its components.

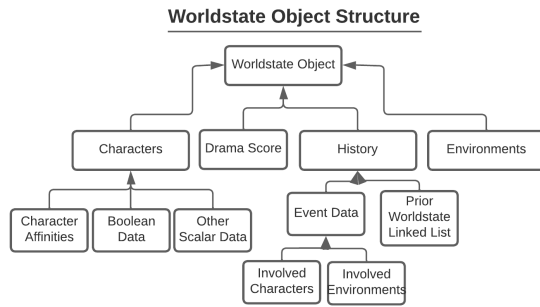


Figure 1: The structure of a Worldstate object and its components.

3.1.2 Waypoints. Waypoints objects are a special case of worldstate objects that are used as targets for the pathfinding system. Whereas a worldstate object is fully instantiated, a waypoint and the characters within may have undefined values for some attributes. This is because when an author is specifying a target for the system, there may be character attributes that they may be indifferent to. In these instances, there is no need to challenge the pathfinding system to find a worldstate with a specific value for that character attribute, and so when the pathfinding system is attempting to find a path that minimizes edit distance between the initial worldstate and the target worldstate, any attributes with undefined values in the target are ignored. Waypoints within our system are also given an author-specified waypoint radius, a tolerance in distance (as calculated by our heuristic) to be met before they are considered satisfied. This is necessary, as it may be that a given waypoint may have target values in character attributes that are relatively prime to the values by which they may be manipulated in the library of plot events, leaving them unreachable. A slack variable allows story generation to progress from waypoint to waypoint without getting hung up on precisely hitting target values.

3.1.3 Characters. Character objects consist of variables that encode attributes individual characters can have, from integer ranges of health and happiness levels, to Booleans encoding binary information about specific characters, such as fugitive or employment status, or other identifying information such as their name. Characters also each have a relationship affinity dictionary/data table that contains a numeric representation of their directional relationship with other characters in the story, tracking how characters perceive one another. Having high levels of granularity over characters is key, as characters are only willing to undergo specific actions, and therefore partake in certain plot events, if their relationships with other characters meet certain prerequisite thresholds.

3.1.4 Plot Fragments. Plot Fragments are events that can take place within a story, and each generated tale is produced utilizing a library of plot fragments that can be selected to run at given points in the story. Each has its own prerequisite checking function, so that it can be determined if a given plot fragment can take place given a current worldstate, and which characters and environments can

be involved. This is fairly consistent with prior efforts in static story generation [20]. Each plot fragment, when run, manipulates attributes of the worldstate and the characters within, and returns an updated worldstate. This involves manipulating the event history to record that a certain plot fragment has been run and what was involved, updating the linked list of prior worldstates, and updating the drama score for the worldstate. All plot fragments have an inherent dramatic value, used to increment that of the updated worldstate, so that drama levels can be tracked across the duration of a generated story. Dramatic values for a given plot fragment are manually tagged in order to demarcate the significance of the event. In our system this ranges anywhere from +20 to -20, with positive values indicating an increase and negative values denoting a decrease in narrative tension. In this manner, we aim to combine some of the aspects of modern drama management systems [19] into our story generation process.

To clarify how plot fragments operate, consider a potential "Murder" plot fragment. A prerequisite checking function for such a plot fragment would ensure that, given a pair of characters 'A' and 'B', 'A' has a strong distaste for 'B' (a motive), a murder weapon in their possession, and that 'A' is proximate to 'B'. The post-conditions of such an event might mark the victim as dead, flag the culprit as a murderer, etc. Fig. 2 depicts such a plot fragment and how it functions.

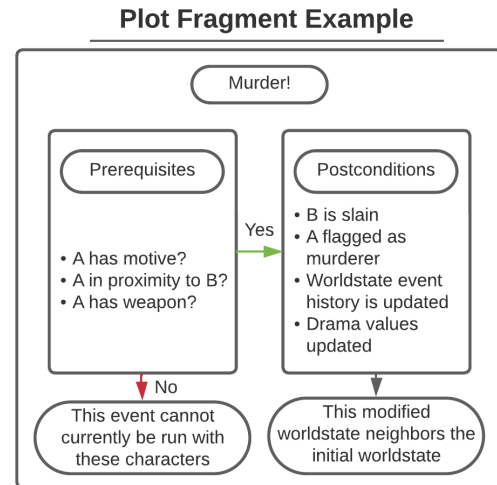


Figure 2: A plot fragment uses a prerequisite checking function to ensure it only occurs in an appropriate story context. If it is, it mutates the initial worldstate, producing a neighboring worldstate node.

Complex prerequisites for plot events can give powerful control over when an event can take place and can be used to ensure characters only act in reasonable manners given their current state. Having characters participate in plot events only when it is rational for them to do so is important for character believability [1].

Systems like Glaive [21] reason about character intentionality in order to keep generated stories coherent, and careful precondition configuration in our system is how we accomplish the same task. By ensuring our characters only partake in actions when it is reasonable for them to do so, we maintain a sense of plausibility. Thus, comprehensive prerequisite checks are critical to ensuring that produced stories are believable.

Finally, plot fragments have associated story text output. When provided with characters and other relevant information, a plot fragment will assemble the completed story text fragment with the story context it has been given. If Alfred and Jean are valid characters for the "HitOnAccepted" event, then

"A flirts with B, and B blushes."

becomes:

"Alfred flirts with Jean, and Jean blushes."

Once a satisfactory story is completed by the system, the system will output all of the associated story text of the plot fragments composing the story to produce a written narrative.

3.2 Pathfinding

3.2.1 Pathfinding Algorithm. In order to drive generated stories toward author-mandated goals, our system treats story generation as a form of pathfinding problem in a graph of possible stories. The graph is relatively easy to characterize: given a starting worldstate node, one can produce a list of all possible variations of events that are permissible to be run, each of which with its own set of postconditions that modify the worldstate. Applying all these postconditions individually to the starting node, we can generate all worldstates reachable within a single plot fragment from the starting worldstate. Without loss of generality, we could continue this recursively on all generated leaf nodes in order to characterize a graph that is composed of all stories possible, given a starting worldstate node. Clearly though, this results in an exponential explosion of nodes, and quickly becomes infeasible to store or traverse. The process of finding neighboring worldstates is outlined below in Fig. 3.

In an effort to blend efficient deterministic methods with a degree of nondeterministic pathfinding to maintain variety in generated stories while reaching author-specified waypoints, we utilize a modified A* [5] algorithm with an element of stochastic story space exploration in conjunction with a distance heuristic that estimates the distance between a given worldstate and a current target waypoint. To put it simply, it is an A* algorithm that with probability 1-p, chooses a random node in the open set rather than selecting the best node in the open set as evaluated by our heuristic. Additionally, as we are not guaranteed to find a path, we store and update the best found path (a node and its associated history) as evaluated by our heuristic while we explore the story graph, and keep a running tally of explored nodes as the system runs. If all states are exhausted or we exceed a user-specified number of explored nodes without meeting all author goals, we return the best found path. If a waypoint is satisfied by coming within a user-specified heuristic tolerance of a waypoint, the open set and closed set are cleared and pathfinding resumes from the node that satisfied the current waypoint towards the next target waypoint, until all targets are

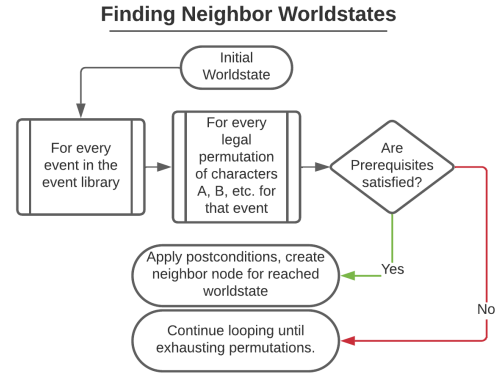


Figure 3: Finding neighbors for a worldstate node involves examining all possible events from a starting state, and all possible worlds reachable from that starting state via those events.

satisfied. The algorithm, in greater detail, can be viewed in Fig. 4 on the following page.

Maintaining a count of explored nodes is important to the system's functionality, as the branching factor for our system can at times exceed 100 with a plot fragment library of less than 40 unique events, and 3 characters. The potential story space is so dense that the system typically cannot explore all possible outcomes, so an author must decide upon an exploration cap for the system. Storing and producing the best result for the author ensures that even if the system fails to meet all author goals, the best approximation found is still provided, and the system indicates that there were issues meeting the waypoints in the time allotted. This collaboration between the system and author helps keep the human in the loop as to what the system is capable of achieving, and can assist the author in recognizing problem areas in their desired waypoints. This provides the author with an opportunity to adjust the provided waypoints or consider the types of plot fragments that might need to be supplied to the system in order to address the desired narrative goals.

The stochastic story space exploration we perform is done for two key reasons:

1. A completely deterministic methodology will always lead to the same output on a given set of inputs. This is contrary to the goals of the system to aid in story generation and ideation. We want variety in output.
2. A completely deterministic approach is liable to become stuck in local minima

While point 1. is self-explanatory, 2. requires further explanation. With a deterministic methodology, our approach can behave similarly to gradient descent in machine learning problems. While still remaining effective under some narrative tasks, as gradient descent is on convex error surfaces, on other tasks, the system may converge to a local minimum without meeting author goals, and continue to churn with little end in sight as a result of the massive

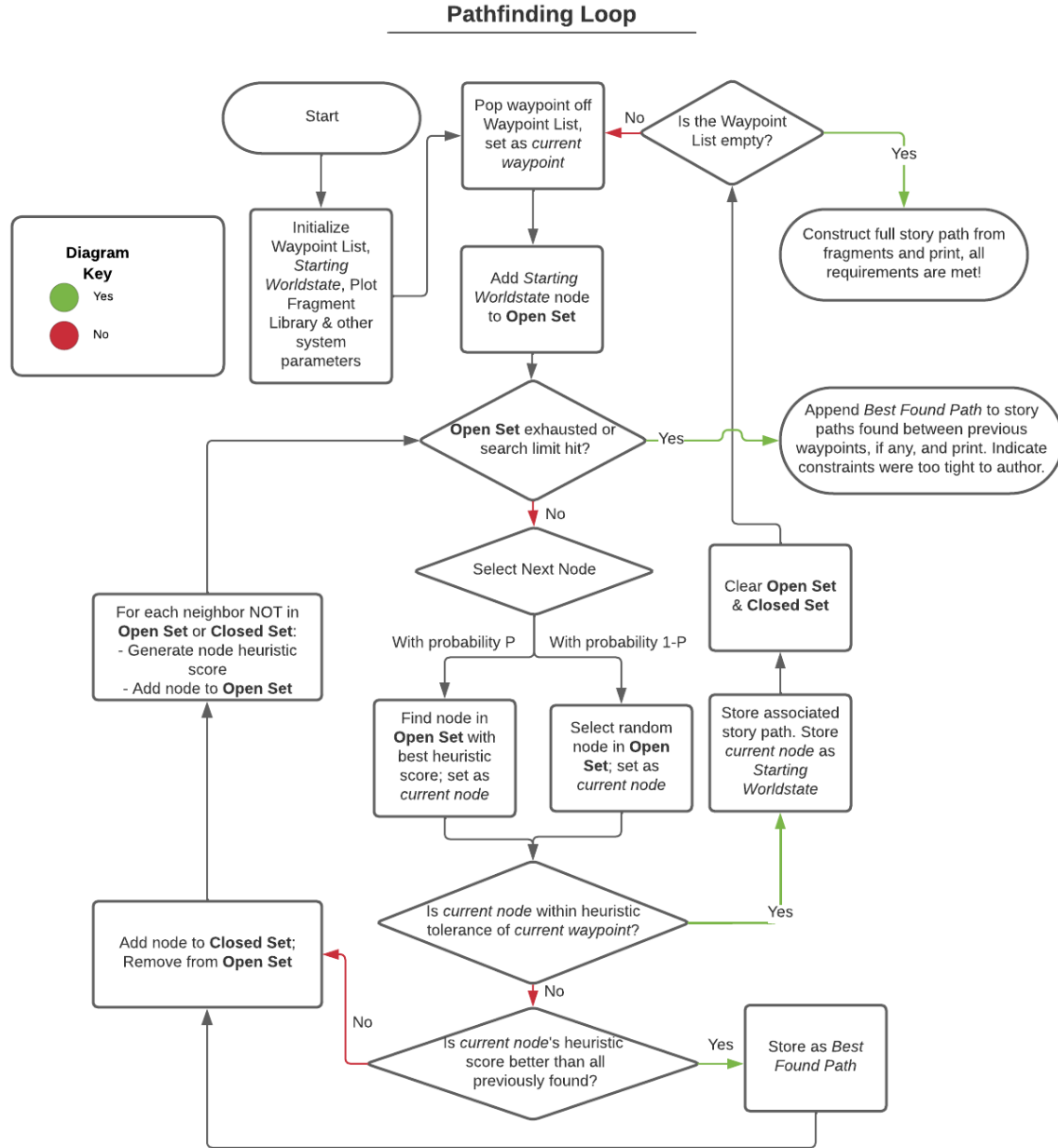


Figure 4: An overview of College Ruled’s pathfinding algorithm.

size of the narrative space. This problem reared its head in early testing. This is parallel to using gradient descent on a task with a concave error surface, where it is guaranteed to converge to a local, rather than global, minimum. Introducing stochastic exploration into both gradient descent and our narrative task here, aids in escaping local minima[6]—our system succeeded in finding story routes that solved narrative tasks that it previously failed to do in a reasonable timeframe.

Stochastic exploration does come at a computation speed cost in scenarios where the deterministic approach would be successful, however. Fig. 5 below plots the count of explored worldstates before finding a solution that met all author goals against P values, where P indicates the probability that the next explored state in the open set is selected deterministically. Notably, some instances of stochastic exploration did result in quicker arrival at a solution for a p -value of 0.7, but this appears to be an exception rather than the rule. There is

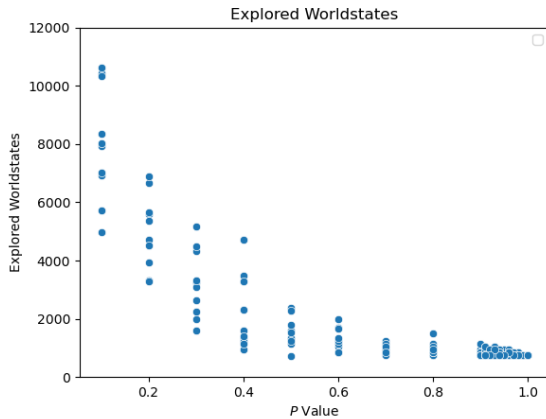


Figure 5: Number of explored worldstates before meeting targets on a story generation task plotted vs p-value, the probability of deterministically selecting the next state to examine with system.

a trade-off for authors to weigh between generating larger variety in output through stochastic exploration, and computation speed.

3.2.2 Pathfinding Heuristic. Our pathfinding heuristic is composed of three separate elements that push story generation toward desired waypoints. Character distance, drama distance, and causality incentives taken together compose our pathfinding heuristic, with each having an important role to play. Character distance or world-state distance calculations push generated tales to meet author goals for character states, drama distance aids in generating stories that match a desirable story tension curve, and causality incentives promote stories with sequences of events that are interrelated.

Combination. Once all three of the previous components—the character distance, drama distance, and causality incentives—are found for a given worldstate, they are combined using a weighted sum with user-specified weights. Thus, an author can focus on generating stories that more closely match their needs. They might care more about matching character attributes to targets than hitting a drama target, and can specify that, while allowing the possibility of freedom to experiment and introduce variety in order areas.

Character Distance. The first component, character distance, is evaluated between a worldstate and a target waypoint by performing a weighted sum over the difference in attributes between the states of characters in the initial state, and their targets in the waypoint. For each parameter associated with a character (be it Boolean, integer, or relationships within their relationship matrix), the difference between the two states is found, transformed using a weighting system that allows for different data types to be evaluated on equal grounds as one another, and then summed. The weights used were all inversely proportional to their range of values. Thus, any two variables, regardless of range of values were considered by the system to be of equal importance. This could be described as a weighted Manhattan edit distance between two worldstates. Prior studies indicate Manhattan edit distance, though

fairly simplistic, remains a fairly effective dissimilarity metric [16]. Characters produced for a desired waypoint are initialized to have all attributes set to "None" or "Undefined" by default. The distance heuristic calculates the distance between any character attribute in the current worldstate and an undefined attribute in the target to be zero. Thus, it only weights and minimizes the distance between attributes the author is actually interested in changing. Creating waypoints is therefore a simple process for authors. In this way, the difference between a worldstate and target can be scored and used to identify worldstates that are closer to a desired target worldstate within a provided waypoint list.

An example of this in action is present in Fig. 6, showcasing a character within an initial worldstate and the same character within a target waypoint with some initialized and undefined attributes. The differences in each attribute are found, transformed via the distance weights, and then summed. Without loss of generality, this procedure is performed across all characters within the initial worldstate and the target waypoint, and all distances are summed until a final distance score is obtained between the current worldstate and the target.

Drama Distance. The second component, drama distance, is evaluated by performing a comparison between a worldstate node and its history with a list of drama targets. The system parses back through the worldstate history of a current worldstate to assemble an ordered list of actual drama values.

Input.

- A current worldstate object denoted W_c and a list of target integer drama values denoted D_t .

Output.

- An integer value denoting the drama distance between the drama history of W_c , denoted D_c , and the target values D_t .

Procedure.

(1) Prior Drama Values Parsing:

- Extract the linked list L of prior worldstate objects from W_c , and initialize the list D_c with the drama score of W_c at its head
 - Traverse the linked list L to extract relevant information from each worldstate object $W_c - 1$ to W_0 .
 - For each object, extract the stored drama value.
 - Prepend the extracted drama values to the list D_c .
- This results in the list D_c being populated with the drama scores of each worldstate including and preceding W_c , in order of occurrence.

(2) Comparison with D_t :

- Once D_c is obtained, compare it with the known list of integer values D_t .
- Sum the differences between corresponding elements in D_c and D_t .

This results in obtaining the summed difference between goals and targets for drama values for the events of the story that have unfolded thus far up to the current story index. Optionally, for each unmatched element in D_t due to differing lengths of D_c and D_t , a flat penalty can be applied to incentivize the system to match the

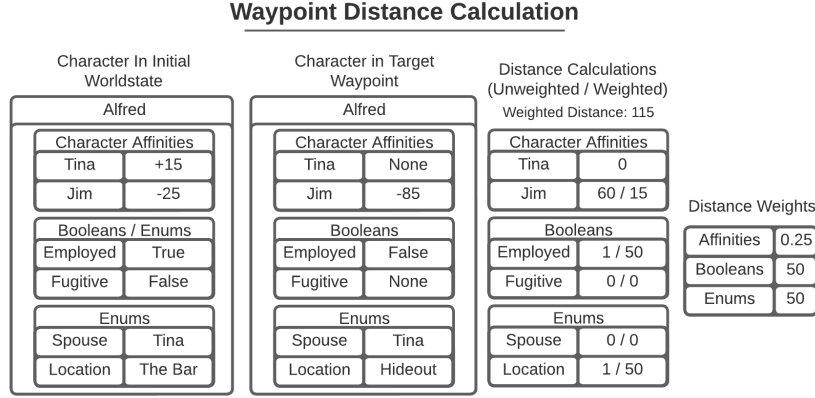


Figure 6: A pair of example worldstates and calculations of the character distance component of the distance heuristic.

full length of desired drama targets instead of concluding the story if other author goals are met early.

Causality Incentive. The final element of our distance heuristic is that of a causality incentive. In order to promote the occurrence of interrelated events in our stories, worldstates with sequences of related events have their calculated pathfinding heuristic reduced depending on the number of causal events in their history.

To determine if a worldstate was produced by a causal event, we examine the circumstances that lead to it’s creation in the following manner:

Definitions.

- **Event:** A specific instance of a plot fragment, with characters, environment, and other parameters determined.

Let us concern ourselves with W_c , the current worldstate, and both W_{c-1} and W_{c-2} – the worldstate immediately preceding W_c , and the worldstate preceding W_{c-1} respectively.

Let E_{c-1} be the set of events runnable from W_{c-1} , and E_{c-2} be the set of events runnable from W_{c-2} .

Let e_{c-1} be the event in E_{c-1} that produced W_c from W_{c-1} :

- If $e_{c-1} \notin E_{c-2}$, then W_c is considered causal.

In plain English: If a worldstate was produced by an event that only became eligible to run after the event preceding it occurred, than we consider that worldstate to be causal.

This is because the event that produced W_c , e_{c-1} , must therefore have only become possible to run after the occurrence of e_{c-2} in E_{c-2} , as it wasn’t present in E_{c-2} . Therefore, e_{c-2} produced a worldstate that allowed e_{c-1} to run, causing it. Therefore, W_c is considered causal.

Given this procedure to determine if a worldstate is causal, determining the total number of causal states in the history of a worldstate object is merely a matter of parsing through its history and utilizing this procedure on each worldstate object, so long as it has at least two preceding worldstates, and keeping a tally. For the

sake of efficiency, we store the number of preceding causal events in each worldstate object as they are generated.

The number of causal events in the history of a worldstate object is referred to as its causality score.

4 ANALYSIS

4.1 Example Tale Analysis

To give a demonstration of the drama management present in our system and the story generation in action we include examples of two stories that were generated with the same waypoints, but with differing drama target curves. In both examples, a story was generated with a series of waypoints that involved the character Mal dating Jess, only to later end up dating Inara, and with a worsened relationship between Jess and both Mal and Inara.¹ This first story was generated with a lower desired drama level:

Mal approaches Jess and strikes up a friendly conversation. They quickly hit it off, sharing stories and interests. Mal mentions their love for cooking and Jess excitedly talks about their recent experiences in traveling. Mal compliments Jess on their sense of humor, and Jess appreciates Mal’s warm personality. Jess asks Mal to go on a picnic with them at the Rocket Wreck Hills. Their heart is racing. Mal blushes and says they would love to go on a date with Jess. Inara warmly approaches Mal and strikes up a heartfelt conversation. They quickly become best friends, sharing stories and laughter. Inara shares a childhood memory, and Mal responds with a funny anecdote from their own past. They find common interests in hobbies and discover that they both love baking. Inara praises Mal’s sense of humor, and Mal admires Inara’s kindness. Mal asks Inara to go on a picnic with them at the Rocket Wreck Hills. Their heart is racing. Inara blushes and says

¹Other unfiltered produced stories can be viewed in the GitHub repository attached in the appendix

they would love to go on a date with Mal. Jess finds out what Mal did and is devastated. They break up with Mal.

In contrast, the following story was generated with the same waypoints, but a higher drama requirement:

Jess warmly approaches Mal and strikes up a heartfelt conversation. They quickly become best friends, sharing stories and laughter. Jess shares a childhood memory, and Mal responds with a funny anecdote from their own past. They find common interests in hobbies and discover that they both love baking. Jess praises Mal's sense of humor, and Mal admires Jess's kindness. Jess asks Mal to go on a picnic with them at the Rocket Wreck Hills. Their heart is racing. Mal blushes and says they would love to go on a date with Jess. Mal asks Inara to go on a picnic with them at the Rocket Wreck Hills. Their heart is racing. Inara declines Mal's invitation. Jess finds out what Mal did and is devastated. They break up with Mal. Jess notices Inara forgot to lock up when they left their bunker. Jess breaks in and steals all their valuables. The law finally caught up with Jess. They are in space jail. Jess spent months chipping at a crack in the circuit panel. They finally succeed at shutting down space jail long enough to make a break for it. Jess returns to their home planet, Higgins. Inara warmly approaches Mal and strikes up a heartfelt conversation. They quickly become best friends, sharing stories and laughter. Inara shares a childhood memory, and Mal responds with a funny anecdote from their own past. They find common interests in hobbies and discover that they both love baking. Inara praises Mal's sense of humor, and Mal admires Inara's kindness. Mal asks Inara to go on a picnic with them at the Rocket Wreck Hills. Their heart is racing. Inara blushes and says they would love to go on a date with Mal.

Although they hit the same beats—two characters initially falling in love, then one cheating and running off with someone else—each plays out differently. The first approaches the same topic in a more reserved manner as a love triangle forms, while the second has more dramatic flair. Following a failed attempt by Mal to cheat on Jess, Jess opts to exact vengeance on the object of her former lover's attention, stealing from the homewrecker Inara, and ending up jailed for her petty attempts at retribution. She ends up jailed for her crimes, before attempting a daring escape. With Jess now out of the picture, Inara and Mal's romance now blossoms.

In both narratives, the same characters fall in love, and the first formed relationship falls apart due to infidelity. However, author-mandated drama levels cause actions and reactions to differ, demonstrating the ability of our drama management system to impact story generation positively. Also, in both instances the author's story goals have been satisfied, ensuring that the drama management is done without compromising other areas of story generation. There are also sequences of causal events which keep the story moving in a coherent manner: In the second story, flirting is followed by dating, then cheating, which results in retribution, then detainment,

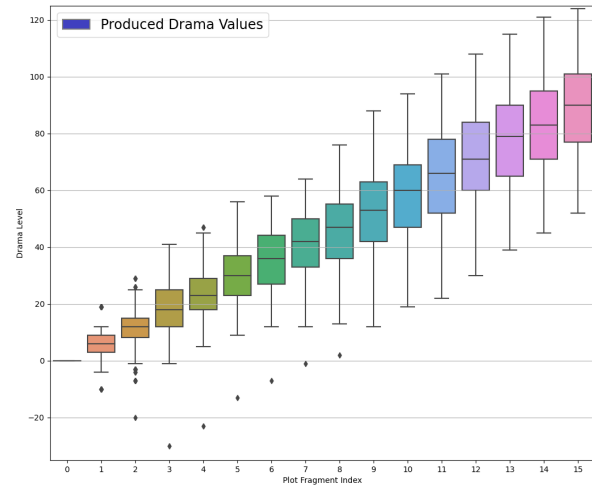


Figure 7: Drama values from 256 stories produced via a random walk on our plot fragment library, as a baseline.

then jailbreak. A sequence of six events, with each of the latter five only made possible by the preceding event. We can see that the causality system is working as intended.

4.2 Graphical Analysis

The box and whisker plot in Fig. 7 charts the spread of drama scores against story index from 256 unique stories generated by performing a random walk on a plot fragment library composed of 36 plot fragments. Fig. 8 showcases generated drama scores produced using the same library of events, but given the same waypoint targets as used in the preceding section to generate stories. While the spread of generated drama values does not match perfectly with the desired target values, it approximates given targets well while generating variety in output and meeting other authorial goals.

A limitation of our system is that it is possible for an author to specify waypoints or a dramatic curve that cannot be reached through provided plot fragments. If an author were to provide a target drama curve that rode above the uppermost regions of reachable drama values in Fig. 7, it might struggle to accommodate such a request. This highlights the importance of a deep library of plot fragments with rich variety in order to provide a robust system for use. A deeper library of events will allow the system opportunities for drama management and manipulation of character attributes to greater extremes. Thus, our system has a degree of overhead cost in terms of generating these plot fragments in order to function at its peak.

5 CONCLUSION

In this paper, we describe College Ruled, a mixed-initiative story generation system, and demonstrate the heuristic-based intelligent pathfinding system to have flexibility in story generation, while prioritizing authorial input and casual sequences of events. College

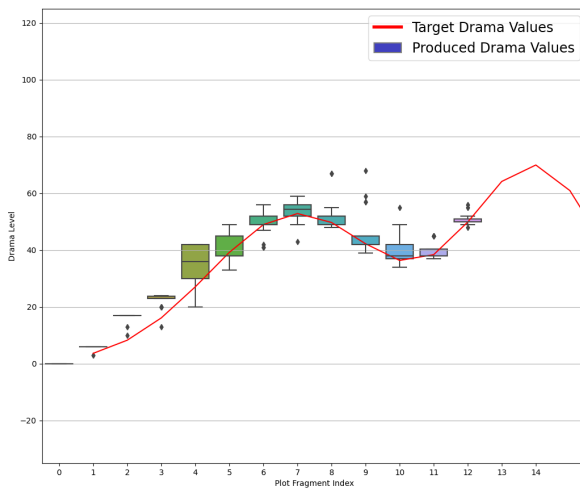


Figure 8: Drama values from 26 stories produced from the same plot fragment library, but given a system of waypoints and a drama curve target. Note relatively tight adherence to target values.

Ruled allows authors to control the dramatic arc of a story in a way not implemented in current static story generators and can be used to explore how a story can achieve the same story beats with different levels of drama, valuable for ideation.

5.1 Future Work

Our current College-Ruled system currently lacks validation by outside sources and has a limited library of authored plot events available for use. Additionally, plot events are not authored by individuals with significant writing experience. Outsourcing the writing of plot events to those with more domain expertise, and performing a user and author study to evaluate the system output remains a high priority. Further exploration of utilizing LLMs to generate plot fragments with greater speed also has the potential to make a system like this much more viable for public use, as one of the greatest constraints of a system like this is the size and quality of the plot fragment library it is provided with. Furthermore, future work on the pathfinding algorithms may be useful. Boolean values can prove troublesome, as there is no gradation in minimizing the distance to a desired Boolean; either the value is true or false. This limits our pathfinder’s ability to incrementally approach world-states that involve changes in Boolean attributes. Implementing Ware et al.’s approach of reasoning backwards from goals in the Glaive narrative planner [21] might prove beneficial in this area in future investigations. Ultimately, we believe this system to be a fruitful effort in bringing new ideas to older author-goal based story generation systems, but believe there is room to expand on our efforts.

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A IMPLEMENTATION

Our codebase, and some example output can be found here:

<https://anonymous.4open.science/r/CollegeRuledFDGAnonymized>.

Received 30 November 2023; revised xx March xxxx; accepted xx June xxxx